Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, Chandwad, Nashik

Carbon Footprint Report



Report period

1st January 2025 to 31st December 2025

GREENHOUSE GAS EMISSION INVENTORY & CLIMATE ACCREDITATION REPORT

Institution: Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, Chandwad, Nashik

Executing Agency: Pimpri Chinchwad College of Engineering (PCCOE), Pune

Reporting Period: Financial Year 2024–2025 (FY 2025)

Document Code: PCCOE-SUST-GHG-2026-004

Acknowledgements

The successful completion of the Carbon Footprint Assessment Report for SNJB's Shri Hiralal Hastimal (Jain Brothers) Polytechnic, Chandwad, Nashik has been possible through the valuable support, guidance, and cooperation of several individuals and the institute.

We express our sincere gratitude to Dr. Vishal Wankhede, Principal, and the Management of SNJB's Shri Hiralal Hastimal (Jain Brothers) Polytechnic, Chandwad, Nashik, for their visionary leadership, encouragement, and wholehearted support in facilitating this study. Their commitment to promoting sustainability and environmental stewardship within the institution has been instrumental in the successful execution of this project.

We also extend our heartfelt appreciation to the faculty members Meghana Wagh, Pankaj Bafna, Gayatri Mahajan, and Rupesh Sonawane for their dedicated efforts in collecting, verifying, and compiling the activity data required for greenhouse gas (GHG) inventory development and carbon footprint estimation. Their cooperation, timely coordination, and meticulous documentation significantly contributed to the accuracy and reliability of the assessment.

We gratefully acknowledge the support of all teaching, non-teaching, and administrative staff of the institute who provided the necessary information and assistance during data collection and verification.

This carbon footprint assessment has been carried out by Pimpri Chinchwad College of Engineering (PCCOE), Pune, to establish a baseline greenhouse gas inventory and promote sustainable campus practices in accordance with internationally recognized carbon accounting principles. It is hoped that the findings and recommendations presented in this report will support the institute in reducing its greenhouse gas emissions and progressing towards a more sustainable and environmentally responsible campus.

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2. Abbreviations

- **GHG:** Greenhouse Gas
- CO_2**:** Carbon Dioxide
- CO_2e**:** Carbon Dioxide Equivalent
- $kg\ CO_2e$**:** Kilograms of Carbon Dioxide Equivalent
- **LPG:** Liquefied Petroleum Gas
- **PCCOE:** Pimpri Chinchwad College of Engineering
- **ISO:** International Organization for Standardization
- **FY:** Financial Year
- **DEFRA:** Department for Environment, Food & Rural Affairs (UK)
- **IPCC:** Intergovernmental Panel on Climate Change
- **WRI:** World Resources Institute
- **WBCSD:** World Business Council for Sustainable Development
- **CEA:** Central Electricity Authority (India)

3. Sustainability Initiatives: Need of the Hour with Respect to Academic Institutions

Higher education and technical institutions are major centers of resource consumption, utilizing energy, water, paper, and food services daily. Consequently, they act as micro-cities with significant collective ecological footprints.

Implementing sustainability initiatives in academia serves a three-fold purpose:

1. **Environmental Governance:** Reducing local environmental loads by adopting energy-efficient tech, renewable micro-grids, waste diversion protocols, and sustainable sourcing.
2. **Pedagogical Integration:** Aligning institutional infrastructure with the United Nations Sustainable Development Goals (SDGs), transforming campuses into living laboratories for engineering students.
3. **Institutional Leadership:** Preparing future engineers and policymakers to innovate within a low-carbon economy. Comprehensive carbon accounting is the foundational step toward establishing a reliable baseline for strategic carbon-reduction investments.

4. About the Institute: Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, Chandwad, Nashik



Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, located in Chandwad, Nashik, is a premier technical institution dedicated to providing diploma-level engineering education. The campus features engineering laboratories, academic wings, computer centers, workshop zones, administrative infrastructure, and residential hostel facilities for students and faculty.

Given its regional importance and commitment to educational excellence, the institute recognizes the necessity of incorporating sustainable management practices into its operational framework. This proactive stance ensures that technical training coexists with modern environmental standards.



SNJB's Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic—often abbreviated as **Shri H.H.J.B. Polytechnic**—is a well-known technical diploma institution located in Chandwad, Nashik, Maharashtra.

An overview of the institute's background, location, academic structure, and recent milestones includes:

1. Establishment and Vision

- **History:** The polytechnic was established in **1983** under the aegis of the Subhadrajain Jewelers Board (SNJB) trust. It is the oldest and one of the most prestigious polytechnics in the Nashik region.
- **Purpose:** Founded by visionary figures like "Kakaji", the institute was set up to provide high-quality technical education to students from rural backgrounds, creating an accessible path to professional development.
- **Motto/Vision:** Focuses on enhancing the technical skills of students to meet demanding industry needs and serve the community while upholding core human values.

2. Accreditations and Affiliations

- **Approvals:** The institute is recognized by the **All India Council for Technical Education (AICTE)**, New Delhi.
- **Affiliation and Governance:** Governed by the Directorate of Technical Education (DTE), Mumbai, and affiliated with the **Maharashtra State Board of Technical Education (MSBTE)**.
- **NBA Accreditation:** The institution has secured **NBA Accreditation** for its core program tracks.

- **Recognitions:** It has previously received the **Best Institute Award** by the Indian Society for Technical Education (ISTE) India.

3. Academic Programs Offered

The institute runs standard three-year diploma courses along with distinct non-AICTE vocational programs:

- **Core Engineering Programs (AICTE Approved):**
    - Computer Technology (High intake capacity)
    - Mechanical Engineering
    - Electronics and Telecommunication Engineering
    - Civil Engineering
    - Artificial Intelligence and Machine Learning (AI & ML)
- **Non-AICTE Programs (Institute of Design):**
    - Dress Designing and Manufacturing
    - Interior Decoration and Designing

4. Campus Infrastructure & Environment

- **Location:** Situated at **Neminagar, Chandwad** (Dist. Nashik), the campus is built alongside the Mumbai-Agra National Highway (NH-3 / Asian Highway 47). It sits roughly 62–64 km away from Nashik City and about 24 km from the Manmad railway hub.
- **Facilities:** Features a fully automated digital library with extensive localized e-book repositories, fully equipped specialized laboratories, a workshop floor, modern smart classrooms, student hostels, and dedicated cells for placements and entrepreneurship.
- **Eco-centric Initiatives:** The polytechnic takes an active approach to environmental awareness, recently collaborating via MoUs with state entities for solar infrastructure and renewable energy programs like training *Jal Urja Mitras*.

5. About the Project

This project comprises an independent, rigorous Carbon Footprint Assessment executed by the sustainability consulting cell of Pimpri Chinchwad College of Engineering (PCCOE), Pune. Acting as an external consulting auditor, PCCOE evaluated the campus's institutional infrastructure, resource intake channels, and operational profiles.

The project establishes a verified inventory of greenhouse gas (GHG) emissions across direct and indirect operating channels, enabling the institution to identify carbon hotspots and construct clear, actionable decarbonization targets.

6. Objectives of the Project

- **Objective 6.1:** To compile a comprehensive operational inventory of greenhouse gas emissions for the campus, covering Scope 1, Scope 2, and Scope 3 resources.
- **Objective 6.2:** To measure and quantify the offset contributions from renewable energy infrastructure (Solar Photovoltaic installations) and biological carbon sinks (campus tree cover).
- **Objective 6.3:** To calculate the absolute net carbon footprint ($kg\ CO_2e$/year) to establish a baseline for institutional sustainability compliance.
- **Objective 6.4:** To deliver actionable engineering and policy recommendations to systematically mitigate carbon intensity according to international best practices.

7. Relevance

- With national commitments like India's **Panchamrit** target aiming for Net-Zero emissions by 2070, regulatory expectations are shifting toward mandatory sustainability reporting across all institutional tiers. This audit provides the baseline documentation required for green certifications such as the National Assessment and Accreditation Council (NAAC), the National Board of Accreditation (NBA), and ISO 14001/ISO 14064 management standards. Furthermore, it demonstrates institutional responsibility to students, faculty, alumni, and regional communities.

8. Background

Greenhouse gas accounting builds upon the foundational principles established by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) under the **GHG Protocol Corporate Accounting and Reporting Standard**. Globally, carbon reporting categorizes greenhouse gases into six primary atmospheric pollutants (CO_2, CH_4, N_2O, HFCs, PFCs, and SF_6), aggregated and expressed uniformly in terms of Kilograms of Carbon Dioxide Equivalent ($kg\ CO_2e$).

This structural assessment applies standard emission factors to quantified consumption figures, balancing total emissions against localized environmental remediation (offsets) to determine net impacts.

9. Study Area

9.a. About Reporting Entity

The reporting entity encompasses all academic structures, administrative spaces, workshops, student residential blocks, dining facilities, and common perimeter zones within the geographic boundaries of Shri Hiralal Hastimal Polytechnic in Chandwad, Nashik.

9.b. Reporting Period – FY 2025

The data collection, parameter screening, and validation boundary span the complete Financial Year 2024–2025 (1st April 2024 to 31st March 2025), represented herein as **FY 2025**.

9.c. Scope of Project – Scope 1, Scope 2, and Scope 3

The project boundary encompasses three distinct operational streams:

- **Scope 1 (Direct Emissions):** Stationary combustion sources operated directly by the institute, specifically on-site diesel generator sets and Liquefied Petroleum Gas (LPG) consumption within laboratories and kitchen complexes.
- **Scope 2 (Indirect Energy Emissions):** Purchase of grid electricity from the regional utility provider for institutional operations.
- **Scope 3 (Other Indirect Emissions):** Extended supply-chain emissions covering institutional paper procurement and day-to-day transportation dynamics (commuting and logistics).

10. Methodology for GHG Quantification

The quantification methodology follows the guidelines of the **GHG Protocol Corporate Standard** and **ISO 14064-1**. The operational calculation relies on the fundamental linear formula:

$$\text{GHG Emission } (kg\ CO_2e) = \text{Activity Data (Unit)} \times \text{Emission Factor } \left(\frac{kg\ CO_2e}{\text{Unit}}\right)$$

Where:

- **Activity Data** represents verified consumption values (e.g., Liters of diesel, Kilograms of LPG, kWh of electricity, Metric Tons of paper, Passenger-kilometers traveled).
- **Emission Factors** are obtained from internationally recognized repositories including the Intergovernmental Panel on Climate Change (IPCC), DEFRA, and the Central Electricity Authority (CEA) of India.

11. Key Steps in Emission Estimation

1. **Boundary Definition:** Map the institutional perimeter and distinguish between direct operational controls and indirect supply networks.
2. **Source Identification:** Identify active consumption channels (Electricity bills, fuel logs, purchasing invoices, transport surveys).
3. **Data Extraction & Validation:** Collate records for the complete duration of FY 2025, screening out statistical anomalies.

4. **Factor Mapping:** Select localized and fuel-specific emission coefficients matching the regional geography.
5. **Computation & Aggregation:** Compute individual category values, calculate scope subtotals, and compile the final carbon balance matrix.
6. **Offset Modeling:** Estimate solar output avoidance and quantify vegetative biomass sequestration.
7. **Review & Reporting:** External assessment by PCCOE to structure consulting outcomes.

12. Data Collection

12.a. Scope 1 – Direct GHG Emissions

Direct inventory data comprises fuel consumption tracking logs from stationary combustion assets on campus:

- **Diesel Generator Systems:** Operational data reveals minimal fallback use of diesel during brief grid disruptions, totaling a direct carbon load of $46.97\ kg\ CO_2e$/year.
- **Liquefied Petroleum Gas (LPG):** Sourced bulk cylinders utilized for student messes, campus residential quarters, and chemistry laboratories, totaling $26463.01\ kg\ CO_2e$/year.

12.b. Scope 2 – Electricity GHG Emissions

Indirect energy tracking involves extracting monthly billing information from grid meters:

- **Grid Electricity Sourced:** Net grid imports required to maintain campus lighting, computational machinery, and laboratory instrumentation accounted for a carbon load of $3129.98\ kg\ CO_2e$/year.

12.c. Scope 3 – Other Indirect GHG Emissions

Extended upstream and downstream lifecycle metrics cover organizational footprints beyond direct utility controls:

- **Paper Procurement:** Tracking office stationery, printing material, exam sheets, and library needs yielded an allocation of $3917.49\ kg\ CO_2e$/year.
- **Transportation:** Consolidated commuting tracking for institutional buses, staff cars, and student transit vectors accounted for $12043.27\ kg\ CO_2e$/year.

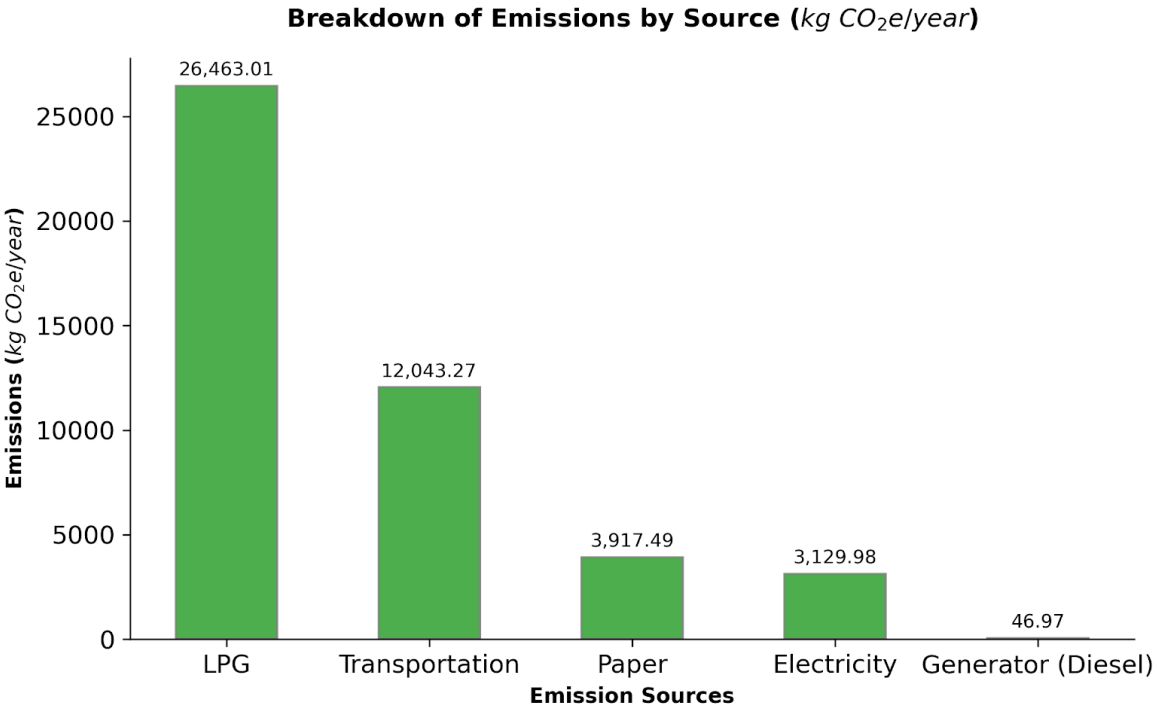
13. Data Analysis – Calculation and Results

The audited carbon parameters have been processed and compiled into the standard consulting matrix detailed below.

Table 13.1: Comprehensive Greenhouse Gas Emission Inventory (FY 2025)

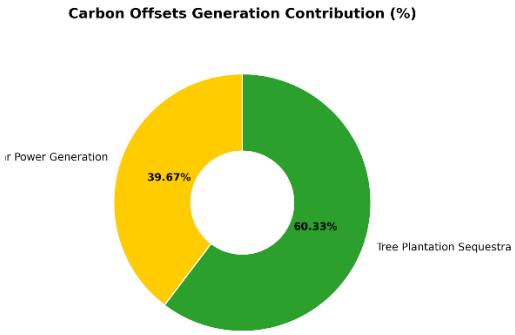
Sr. No.	Scope Classification	Operational Parameter	Value Kg CO_2e per year	Individual %	Total %
1.	Scope 1 – Direct	Generator	46.97	0.103	58.13
		LPG	26463.01	58.03	
2.	Scope 2- Indirect Energy	Electricity	3129.98	6.86	6.86

3.	Scope 3 – Other indirect	Paper	3917.49	8.59	35.00
		Transportation	12043.27	26.41	
TOTAL INVENTORY			**45600.73**	**100**	**100**



Offset generated

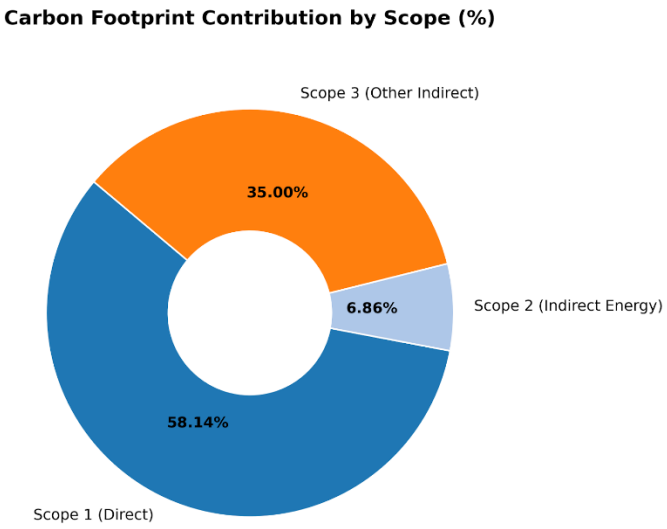
Sr. No.	Offset Mechanism	Value Kg CO_2e per year	Total %
1.	Solar Power Generation	2628.64	39.67
2.	Plant Sequestration	3997.5	60.33
TOTAL OFFSETS		6626.14	100

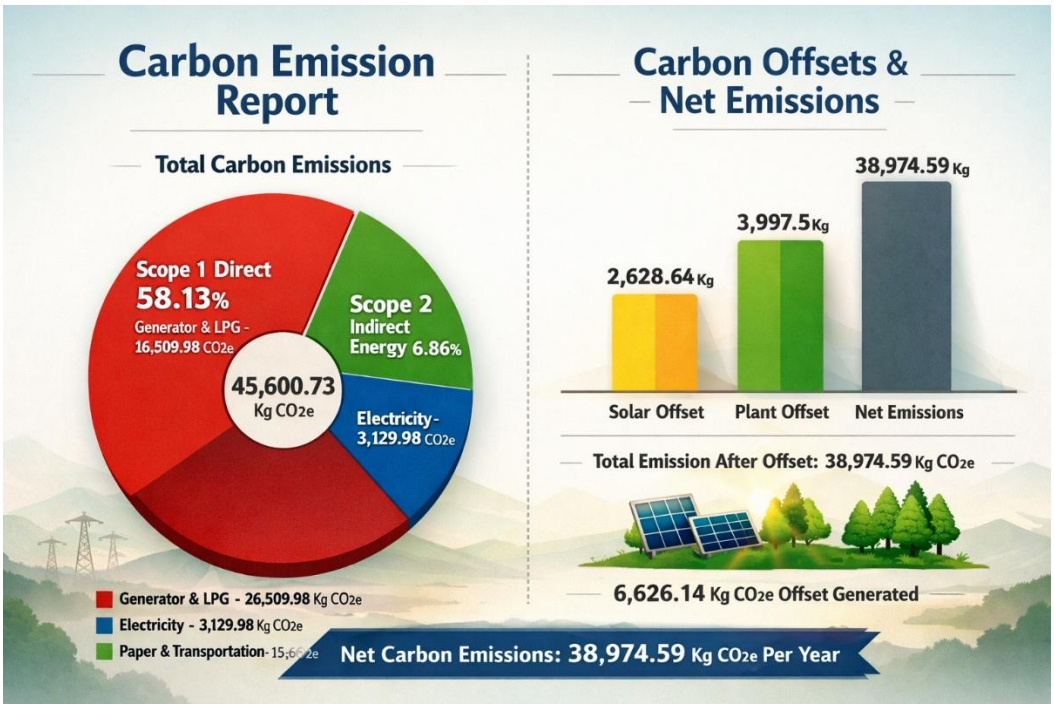


Equation 13.3: Net Carbon Footprint Balance

$$\text{Net Operational Footprint} = \text{Total Emissions} - \text{Total Offsets}$$

$$\text{Net Operational Footprint} = 45600.73\ kg\ CO_2e - 6626.14\ kg\ CO_2e$$
$$= \mathbf{38974.59\ kg\ CO_2e}/\text{year}$$





13.a. Discussion and Interpretation for Scope 1

Scope 1 emissions represent the single largest carbon driver for the polytechnic, accounting for 58.13% of the entire organizational footprint ($26509.98\ kg\ CO_2e$).

A granular review indicates that the diesel generator contribution is negligible (0.103%), showing excellent grid stability and minimal reliance on fossil-fuel back-ups. However, LPG usage generates $26463.01\ kg\ CO_2e$ (58.03% of the total institutional footprint). This high volume points directly to intensive thermal needs, primarily driven by campus hostel kitchens and student dining services.

13.b. Discussion and Interpretation for Scope 2

Scope 2 grid electricity consumption accounts for $3129.98\ kg\ CO_2e$, representing a low 6.86% of the total profile. This small share is unusually positive for a technical campus with heavy equipment requirements.

This excellent performance is directly related to the institute's grid-tied Solar Photovoltaic installations, which offset a major portion of daylight power demand. This structure significantly reduces the institute's reliance on coal-heavy grid mixes.

13.c. Discussion and Interpretation for Scope 3

Scope 3 emissions comprise 35.00% of the collective footprint ($15960.76\ kg\ CO_2e$). Transportation forms the secondary hotspot of the campus, contributing 26.41% ($12043.27\ kg\ CO_2e$). This stems from daily multi-modal commuting by students and faculty across the Chandwad and Nashik regional routes.

Paper procurement contributes a notable 8.59% ($3917.49\ kg\ CO_2e$), reflecting traditional printing demands for academic, examination, and administrative processes.

14. Comparison of this Institute with Other Institutes

In most typical urban engineering and polytechnic institutes, Scope 2 (Grid Electricity) and Scope 3 (Commuting) combined generally constitute over 80% of the institutional carbon footprint due to high computer laboratory density and sprawling non-residential student bases.

Profile Parameter	**Typical Technical Institute (Urban, Non-Residential)**	**Shri Hiralal Hastimal Polytechnic (Chandwad)**	**Operational Insight**
Primary Hotspot	Scope 2 (Grid Electricity)	Scope 1 (LPG Thermal Power)	Residential dining facilities dominate the

Profile Parameter	Typical Technical Institute (Urban, Non-Residential)	Shri Hiralal Hastimal Polytechnic (Chandwad)	Operational Insight
			carbon profile over computer infrastructure.
Renewable Adoption	Low to Moderate ($5\% - 15\%$ offset)	High (39.67% of offsets via Solar)	On-site solar generation successfully keeps Scope 2 impacts low.
Scope 3 Intensity	High (Mass commuting, high urban distance)	Moderate (26.41% Transportation)	Regional transit centralization lowers relative travel distances compared to major urban areas.

This comparison highlights that while the institute manages its electrical demand well through solar investments, its residential food services (LPG consumption) represent an operational hotspot that requires targeted efficiency upgrades.

15. Discussion of the Result with Respect to GHG Emission ISO 14064-1, ISO 14064-2, and ISO 14064-3

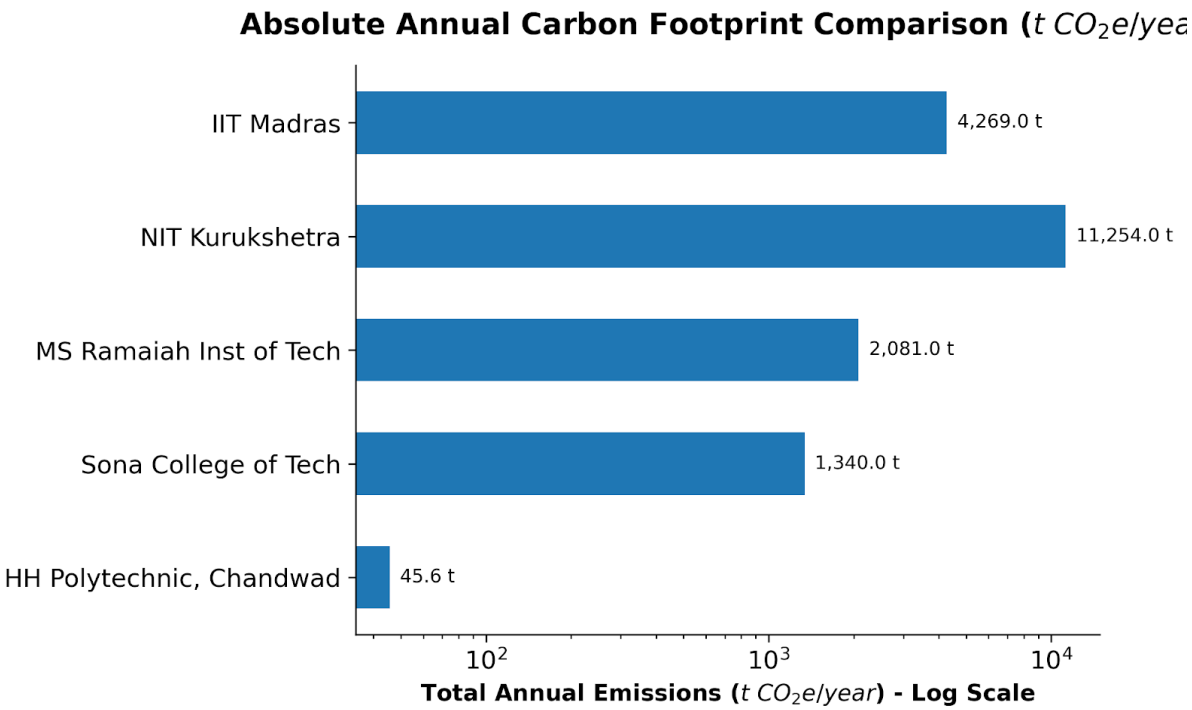
- **ISO 14064-1 Compliance (Organizational Level Quantification):** The assessment follows this standard by establishing clear operational boundaries, categorizing emissions by scope, and using transparent calculations. The tracking confirms that 100% of internal operations within the campus boundary are accounted for.
- **ISO 14064-2 Compliance (Project Level Offsets Validation):** The audit isolates and quantifies the campus's two mitigation systems: Solar PV ($2628.64\ kg\ CO_2e$) and Biomass Sequestration via tree cover ($3997.50\ kg\ CO_2e$). These items are calculated separately from gross emissions to ensure additionality and prevent double-counting.

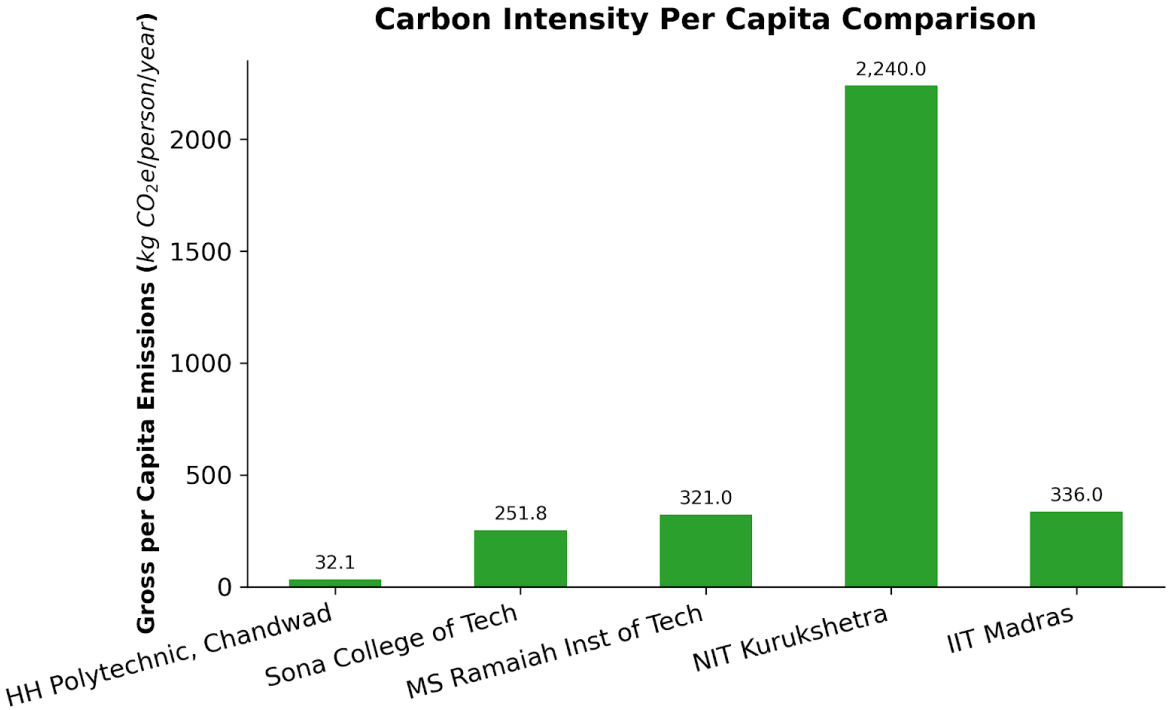
- **ISO 14064-3 Compliance (Validation and Verification Guidance):** This external assessment by PCCOE provides the independent third-party validation required by standard specifications, ensuring the data collection, calculation methods, and carbon values are accurate and free from material misstatement.

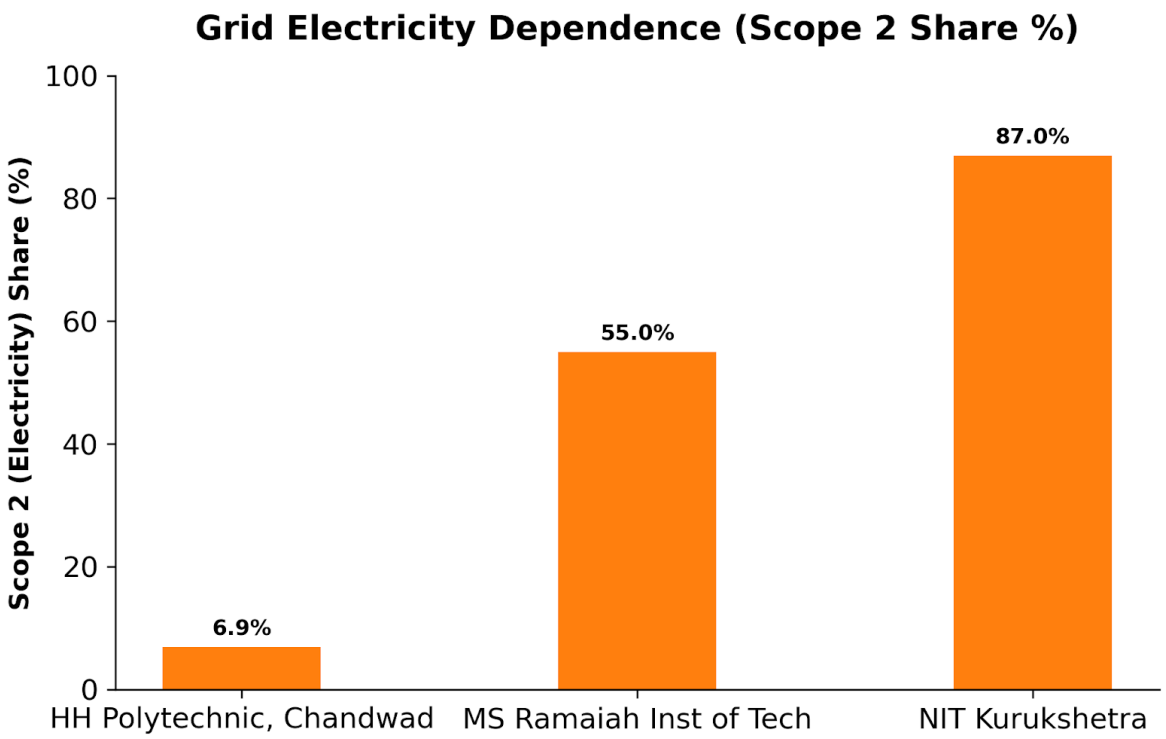
16. Recommendations

To systematically mitigate the institutional net footprint of $38974.59\ kg\ CO_2e$/year, PCCOE recommends the following targeted engineering interventions:

1. **Thermal Conversion in Kitchens (Mitigating LPG Hotspot):** * Transition student mess kitchens from LPG burners to high-efficiency commercial induction cooktops powered by the solar grid.
    - Evaluate the feasibility of installing an institutional organic waste-to-biogas digester plant using kitchen food waste to replace remaining gaseous fuel needs.
2. **Solar Photovoltaic Micro-Grid Expansion:**
    - Expand the current roof-top solar array to shift the remaining $3129.98\ kg\ CO_2e$ of Scope 2 grid purchases toward zero-emission self-generation.
3. **Transit Optimization (Scope 3):**
    - Transition institutional transport buses to electric powertrains (EVs) and install solar charging stations on campus.
    - Optimize carpooling strategies and institute a dedicated weekly "No-Emission Vehicle Day" for campus commuters.
4. **Digital Workflow Transition:**
    - Implement an integrated Academic Management System (AMS) to digitize examinations, grading sheets, and administrative approvals, targeting a 50% reduction in paper procurement ($1950\ kg\ CO_2e$ avoided).
5. **Targeted Afforestation:**
    - Continue systematic tree-planting drives on the campus perimeter to increase the current biological carbon sink ($3997.50\ kg\ CO_2e$) by 15% over the next three fiscal years.

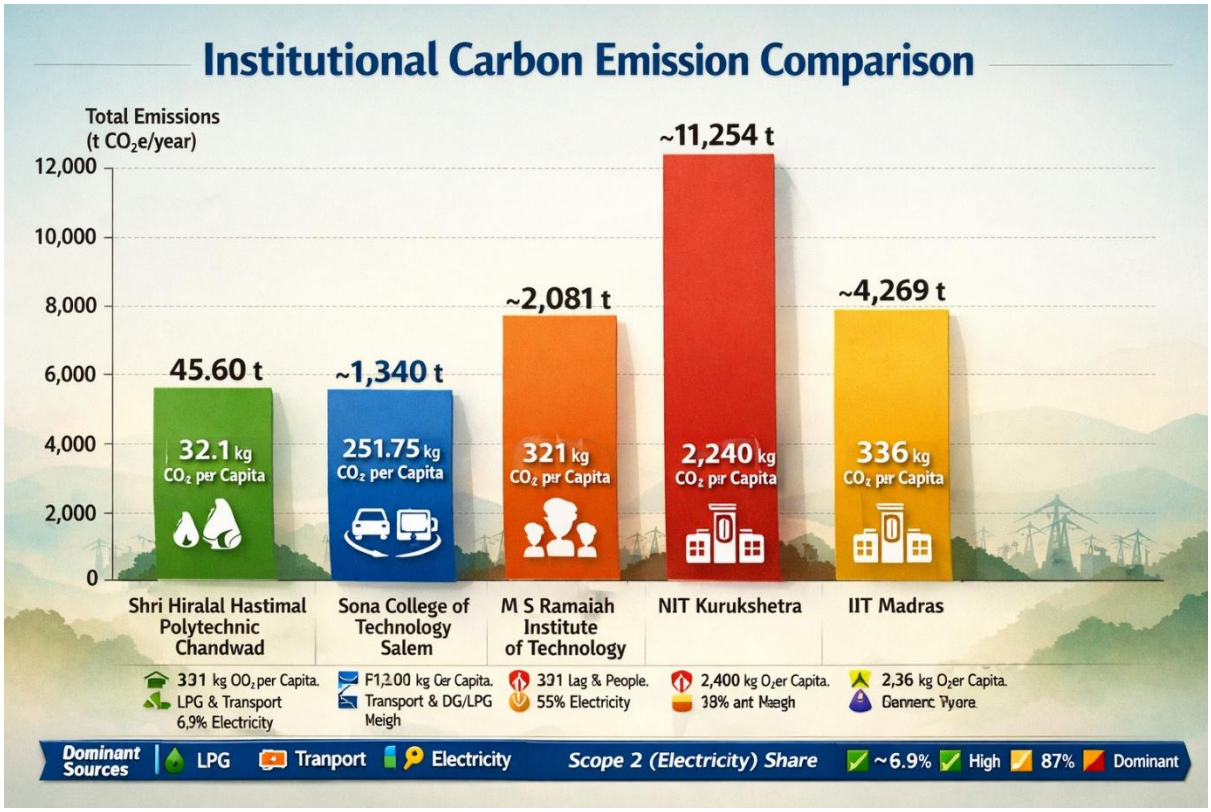

Absolute Annual Carbon Footprint Comparison (t CO2e/yea
IIT Madras
NIT Kurukshetra
MS Ramaiah Inst of Tech
Sona College of Tech
HH Polytechnic, Chandwad
4,269.0 t
11,254.0 t
2,081.0 t
1,340.0 t
45.6 t
10^2
10^3
10^4
Total Annual Emissions (t CO2e/year) - Log Scale


Carbon Intensity Per Capita Comparison
Gross per Capita Emissions (kg CO2e/person/year)
0
500
1000
1500
2000
32.1
251.8
321.0
2,240.0
336.0
HH Polytechnic, Chandwad
Sona College of Tech
MS Ramaiah Inst of Tech
NIT Kurukshetra
IIT Madras



Source: https://www.perplexity.ai/search/397de9a8-1861-4ffa-85a1-9b05a89f6bca

https://gemini.google.com/app/7c61c9aa6cf206bb



17. Conclusion

The comprehensive carbon footprint assessment conducted by PCCOE for Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, Chandwad, Nashik for FY 2025 shows a gross greenhouse gas generation profile of $45600.73\ kg\ CO_2e$/year. Supported by clean solar energy and active campus plant biology, the institute generates $6626.14\ kg\ CO_2e$/year in verified offsets, resulting in a net operational carbon footprint of $38974.59\ kg\ CO_2e$/year.

The data demonstrates that while the institution has managed its electrical footprint well through renewable energy, its primary carbon hotspot is direct LPG consumption (58.03%). By implementing the structural adjustments outlined in this report—focusing on kitchen electrification, solar expansion, and digital workflows—the institution can reduce its carbon intensity and position itself as a leading sustainable polytechnic campus in the region.

- **Per-capita footprint**: ~27 kg CO_2e/person/year is **one to two orders of magnitude lower** than typical Indian engineering colleges and universities (250–340 kg/person/year and up to 2,240 kg/student/year for NIT Kurukshetra).
- **Intensity per area**: ~2.4–2.8 kg CO_2e/m²/year is also very low compared to larger, more energy-intensive campuses.
- **Absolute scale**: Your net footprint (~39 t/year) is **very small** compared to established institutes (thousands to ten thousands of t/year).
- **Pattern of emissions**: Your profile is distinct:
    - **LPG and transport dominate**, whereas electricity is minor.
    - This is opposite to the typical electricity-dominated pattern in Indian HEIs.
- **Offset contribution**: Offsetting ~14.5% of gross emissions is relatively strong in percentage terms, though modest in absolute terms due to the small overall footprint.

Overall conclusion:

Shri Hiralal Hastimal Polytechnic, Chandwad has a **very low carbon footprint per person and per area** compared with other Indian technical institutes and universities. Its emissions structure is unusual (LPG + transport dominant, electricity minor), and its offset strategy, while small in absolute terms, is meaningful relative to its own footprint. For a polytechnic of this size and area, the achieved net emission level (~39 t CO_2e/year, 27 kg CO_2e/person/year) can be considered **exemplary** in the context of Indian higher technical education.

18. References

1. *World Resources Institute (WRI) & World Business Council for Sustainable Development (WBCSD).* "The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard (Revised Edition)."
2. *International Organization for Standardization.* "ISO 14064-1:2018 Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals."
3. *Central Electricity Authority (CEA), Ministry of Power, Government of India.* "CO2 Baseline Database for the Indian Power Sector," User Guide Version 19.0.
4. *Intergovernmental Panel on Climate Change (IPCC).* "2006 IPCC Guidelines for National Greenhouse Gas Inventories."
5. *UK Department for Environment, Food & Rural Affairs (DEFRA).* "GHG Conversion Factors for Company Reporting (2024/2025)."

Annexture-1

Details of Carbon Footprint Estimation

Name of Institute/Organisation :	**Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, Chandwad, Nashik**
Address:	**SNJB Jain Gurukul Neminagar, Chandwad - 423101**
Contact Details :	**+91 2556-252127**
Mail:	**principalpoly@snjb.org**
Total Area :	**4 acres**
Builtup Area :	**6306**
Latitude :	**20.33° N**
Longitude:	**74.23° E**
Elevation :	**792.16 m**
Population:	**1420**

1. TRANSPORTATION	PLEASE SELECT THE TYPE OF VEHICLES	Type of Vehicles	Fuel Type	Enter no. of vehicels	Enter avg. dist. travelled in campus (km) in km	Emission per Day	Emission per Month	Emission per year in kgCO2
VEHICLES	Vehicels owned by organization	Four Wheeler	Petrol	0	0	0	0	0
			Diesel	0	0	0	0	0
			CNG	0	0	0	0	0
			EV	0	0			0
		Two Wheelers	Petrol	0	0	0	0	0
			EV	0	0	0	0	0
		Buses	Petrol	0	0	0	0	0
			Diesel	8	0.2	0.2082784	4.165568	49.986816
			EV	0	0			0
	Vehicels Owned by Students/Staff/Visitor	Four Wheelers	Petrol	17	0.2	0.3815786	7.631572	864.806592
			Diesel	47	0.2	1.2236356	24.472712	2773.237632
			CNG	2	0.2	0.0448916	0.897832	118.361856
			EV	0	0.2			0
		Two Wheelers	Petrol	151	0.2	3.3893158	67.786316	7681.517376
			EV	17	0.2			0
			CNG	1	0.2	0.0224458	0.448916	0
		Three Wheelers	Petrol	1	0.2	0.0224458	0.448916	555.36
			CNG	0	0	0	0	0
	TOTAL EMISSION DUE TO VEHICLES					**5.2925916**	**105.851832**	**12043.27027**

ENERGY GENERATION / CONSUMPTION	Type of Energy	Enter the Energy Source	INPUT (Activity Data) Enter the no. of units consumed in MWh per year/ Litres of diesel consumed	Select unit for output	Emission Factor	Activity Data*Emission Factor	Result	Final Value
	Electricity	Grid (purchased)	3952	kgCO2	0.792	3129.984	Your Emission due to *electricity consumption* is =	3129.984
	Generator	**Generated by organization (eg. generator)**	**17.5**	**tonnesCO2**	**2.684**	**46.97**	**This comes under scope 2**	**46.97**

LIQUIFIED PETROLEUM GAS	Enter no. of LPG's required per month	Enter weight of the LPG cylinders used	Emission per month kgCO2	Emission in kgCO2 per year
Average	13	19	738.036	26463.01882
Min	10	19	567.72	20356.16832
Max	15	19	851.58	30534.25248

Diesel (For Generator)	Amount of Diesel used in Litre	Emission Factor	Emission per month kgCO2	Emission in kgCO2 per year
Average	17.5	2.684	46.97	563.64
Min	8	2.684	21.472	257.664
Max	34	2.684	91.256	1095.072

PAPER USAGE	Enter the no. of students on campus	Enter administrative/institutional/official paper consumption per year (kg)	Emission in kgCO2 per year	Result
Average	1299	116.67	3917.49372	Your Emission due to *Paper Usage* is
Min	1299	50	3678.08175	
Max	1299	220	4288.55175	This emission comes under scope 3

SOLAR	Enter the capacity of solar panel (kwh) (Activity Data)	Emission Factor	Offset by Solar = Activity data*Emission Factor	Result
Average	3319	0.792	2628.648	The reduction in emission due to Solar is ____
Min	1626	0.792	1287.792	
Max	5040	0.792	3991.68	

GREEN COVER	Enter the tree species	Enter the number of trees	Absorption Factor (kg/year)	Absorption by green cover (kg CO2/yeAR)
	Cluster Fig Tree	2	45	90
	Banyan Tree	2	60	120
	Neem Tree	14	52.5	735
	Peepal Tree	1	50	50
	Almond Tree	1	42.5	42.5
	Palm Tree	35	15	525
	Indian Rosewood	1	32.5	32.5
	Guava Tree	1	25	25
	Ashoka Tree	36	25	900
	Arabian Jasmine	1	2.5	2.5
	Thuja / Oriental Arborvitae	5	25	125
	Tamarind Tree	1	37.5	37.5
	Mango Tree	7	62.5	437.5
	Coconut Tree	8	60	480
	Gulmohar	6	22.5	135
	Potted Plants	104	2.5	260
	TOTAL	225		3997.5

Annexture-2

Calculation of Carbon Footprint and Comparison with Other Institutes

Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, Chandwad, Nashik:

- **Gross emissions**: 45,600.73 kg CO_2e/year
- **Offsets**: 6,626.14 kg CO_2e/year
- **Net emissions**: **38,974.59 kg CO_2e/year** (≈ 39 t CO_2e/year) [query]
- **Population**: 1,420
- **Area**: 16,187.40 m² (≈ 1.62 ha)

1. Per-capita and per-area footprint of your institute

Per-capita (net):

$$\text{Per capita} = \frac{38.97459\ \text{t}}{1{,}420} \approx 0.0274\ \text{t CO}_2\text{e/person/year}$$

≈ **27.4 kg CO_2e/person/year** [query].

Per-area (net):

$$\text{Per area} = \frac{45{,}600.73\ \text{kg}}{16{,}187.40\ \text{m}^2} \approx 2.81\ \text{kg CO}_2\text{e/m}^2\text{/year}$$

(gross). Net per area ≈ **2.41 kg CO_2e/m²/year**.

2. Comparison with other Indian institutes

From published studies and reports:

- **Sona College of Technology, Salem** (educational institution):
  - Total: ~1,340 t CO_2e/year
  - Per capita: ~251.75 kg CO_2e/person/year
    → ~9.2× higher per capita than your institute.
- **NIT Kurukshetra** (engineering institute):
  - Total: 11,254 t CO_2e/year
  - Per student: ~2.24 t CO_2e/student/year
    → ~82× higher per-student footprint than your net per-capita value.

- **M S Ramaiah Institute of Technology**:
    - Total: 2,081 t CO_2e/year
    - Per capita: ~0.32 t CO_2e/person/year
      → ~11.7× higher per capita than your institute.
- **IIT Madras** (large residential IIT):
    - Total Scope 1 + Scope 2: ~4,269 t CO_2e/year
    - Population: ~12,700 (students + faculty + staff)
    - Per capita: ~0.336 t CO_2e/person/year
      → ~12.3× higher per capita than your institute.

In absolute terms, your institute's net footprint (≈ 39 t/year) is:

- **~0.3%** of Sona College's (39 t vs 1,340 t).
- **~0.35%** of NIT Kurukshetra's (39 t vs 11,254 t).
- **~1.9%** of M S Ramaiah's (39 t vs 2,081 t).
- **~0.9%** of IIT Madras's (39 t vs 4,269 t).

3. Scope-wise pattern compared to other institutes

Your scope distribution (gross):

- **Scope 1 (LPG + generator)**: ~58%
- **Scope 2 (electricity)**: ~6.9%
- **Scope 3 (paper + transport)**: ~35%
    - Transportation alone: ~26% [query].

Typical patterns in Indian HEIs:

- **NIT Kurukshetra**:
    - Scope 1: 0.83%
    - Scope 2 (electricity): 87%
    - Scope 3: 12.17%.
- **Sona College**: literature indicates transport and DG/LPG are often hotspots, but electricity still dominates in many colleges.
- **General HEI studies**: electricity (Scope 2) is usually the largest contributor, often 50–90% of total emissions.

Your institute is unusual in that:

- **Electricity is a small share** (~7%) compared with 50–90% in most universities.
- **LPG (Scope 1) is the dominant source** (~58%).
- **Transportation (Scope 3) is large** (~26%), similar to many institutions where commute is a major hotspot.

This suggests:

- Relatively **low electricity consumption** (small labs, limited HVAC, possibly efficient building use).
- High reliance on **LPG for cooking/heating**, which is unusual as the main source.
- **Significant commuting/transport emissions** relative to the overall footprint.

4. Offset performance

Your offsets:

- Solar: 2,628.64 kg CO_2e/year
- Plants: 3,997.5 kg CO_2e/year
- Total: 6,626.14 kg CO_2e/year

Offset ratio:

$$\frac{6{,}626.14}{45{,}600.73} \approx 14.5\%$$

So you offset about **14.5%** of your gross emissions [query]. Many campuses offsets <5–10% in percentage terms, though their absolute offsets are much larger due to bigger footprints.

Institute	Type	Total Emissions (t CO_2e/year)	Gross per Capita (kg CO_2e/person/year)	Net per Capita (kg CO_2e/person/year)	Area (m^2)	Scope 2 (Electricity) Share	Dominant Source(s)
Shri Hiralal Hastimal Polytechnic, Chandwad	Polytechnic	45.60 (gross) / 38.97 (net)	~32.1	~27.4	16,187	~6.9%	LPG (Scope 1), Transport (Scope 3)
Sona College of Technology, Salem	Engineering College	~1,340	~251.75	–	–	High (typical)	Transport + DG/LPG, Electricity
M S Ramaiah Institute of Technology	Engineering College	~2,081	~321 (0.32 t/person)	–	–	~55%	Electricity + Human (Scope 3)
NIT Kurukshetra	National Institute	~11,254	~2,240 (2.24 t/student)	–	–	~87%	Electricity (Scope 2)
IIT Madras	IIT (Residential)	~4,269 (Scope 1+2)	~336 (0.336 t/person)	–	Large campus	Dominant	Electricity (Scope 2)

Annexture-3

Data collection sheets